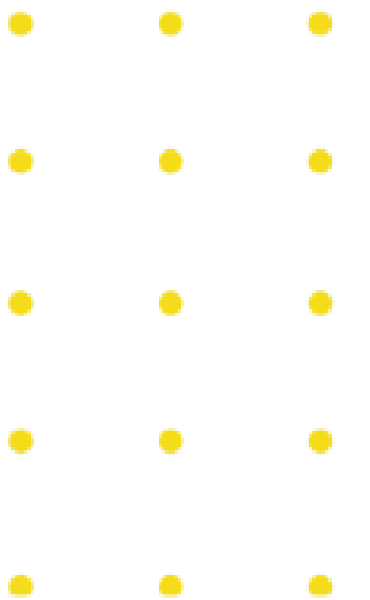
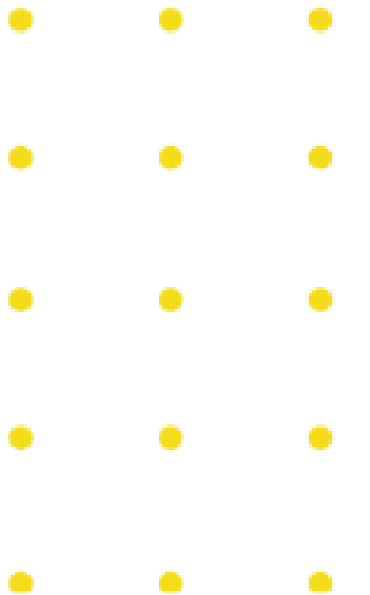


Module 4:

Endpoint Security

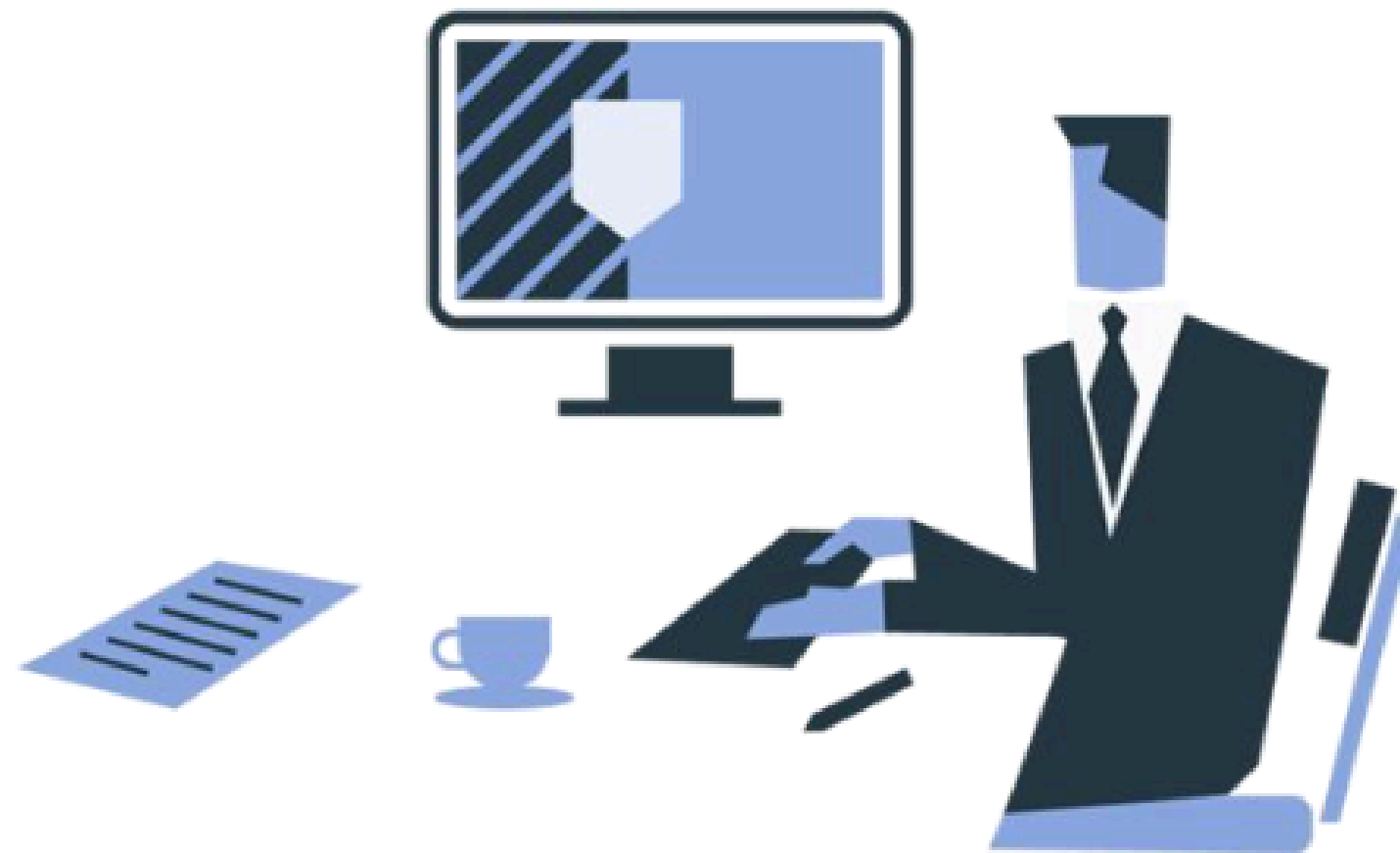


Topic 1: Endpoint Security



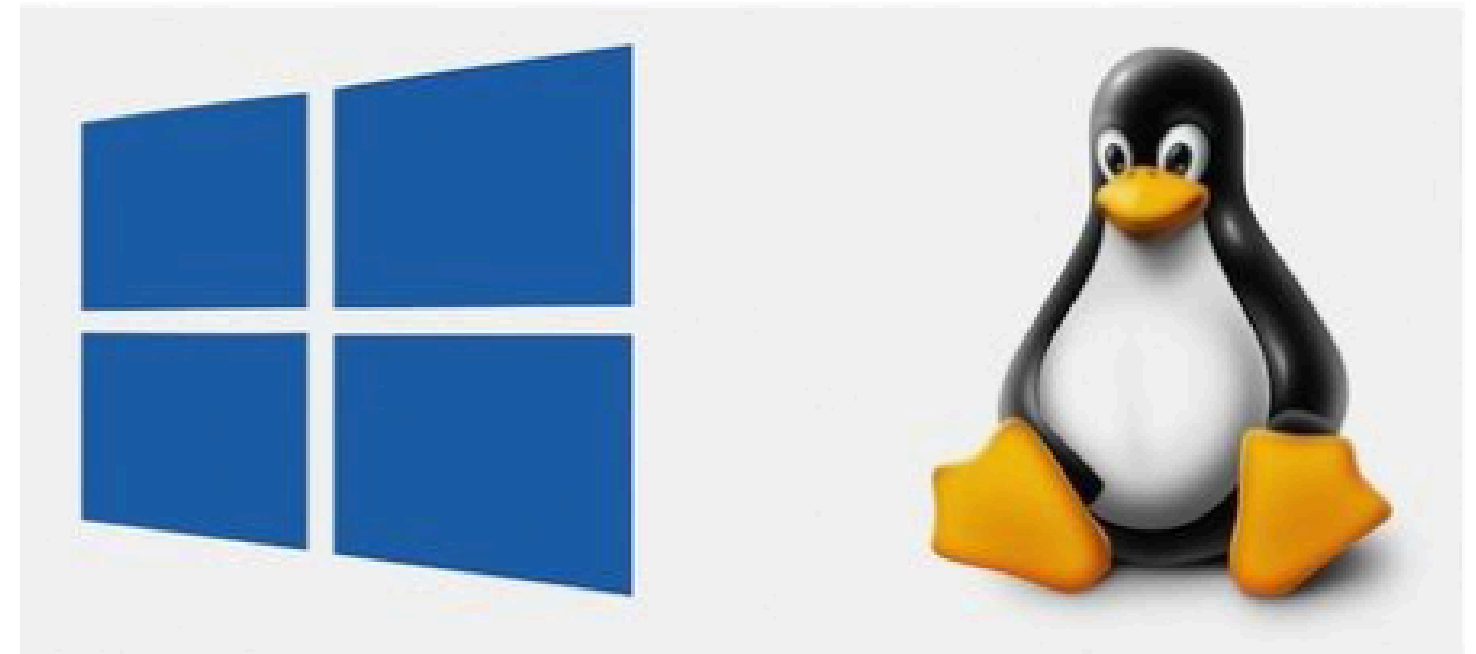
What is Endpoint Security

Endpoint security or entry point security is the actions and procedures taken to secure the points of end-user devices such as laptops, desktop computers, mobile devices for all kinds of cyber threats



Windows vs Linux in Security

- Linux is usually installed on the most important, mission-critical systems
- The most expensive systems are often UNIX or Linux-based
- Most security appliances also run Linux “Firewalls/Routers/IDS”



Endpoint Security Monitoring

- Host intrusion detection system
- Host intrusion prevention system
- Anti-Virus solutions
 - Signature-based
 - Behaviour-based
- Endpoint Host FW (Windows FW/ IPtables)
- Endpoint detection and response

Linux and Windows Processes

C:\WINDOWS\system32\cmd.exe

Microsoft Windows [Version 10.0.17134.112]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\WINDOWS\system32>tasklist

Image Name	PID	Session Name	Session#	Mem Usage
System Idle Process	0	Services	0	8 K
System	4	Services	0	2,692 K
Registry	120	Services	0	6,856 K
smss.exe	644	Services	0	320 K
csrss.exe	752	Services	0	2,316 K
csrss.exe	836	Console	1	3,288 K
wininit.exe	856	Services	0	1,540 K
winlogon.exe	940	Console	1	6,588 K
services.exe	948	Services	0	6,420 K
lsass.exe	984	Services	0	14,940 K
svchost.exe	832	Services	0	1,088 K
svchost.exe	1044	Services	0	18,788 K
UDFHost.exe	1072	Services	0	1,380 K
fontdrvhost.exe	1092	Services	0	984 K
fontdrvhost.exe	1100	Console	1	4,056 K
svchost.exe	1224	Services	0	9,684 K
svchost.exe	1276	Services	0	4,808 K
dun.exe	1336	Console	1	49,088 K

UID	PID	PPID	C	STIME	TTY	TIME	CMD
root	1	0	0	Nov15	?	00:00:03	/sbin/init splash
root	2	0	0	Nov15	?	00:00:00	[kthreadd]
root	3	2	0	Nov15	?	00:00:00	[rcu_gp]
root	4	2	0	Nov15	?	00:00:00	[rcu_par_gp]
root	6	2	0	Nov15	?	00:00:00	[kworker/0:0H-kb]
root	8	2	0	Nov15	?	00:00:00	[nn_percpu_wq]
root	9	2	0	Nov15	?	00:00:00	[ksoftirqd/0]
root	10	2	0	Nov15	?	00:00:02	[rcu_sched]
root	11	2	0	Nov15	?	00:00:00	[migration/0]
root	12	2	0	Nov15	?	00:00:00	[idle_inject/0]
root	14	2	0	Nov15	?	00:00:00	[cpuhp/0]
root	15	2	0	Nov15	?	00:00:00	[cpuhp/1]
root	16	2	0	Nov15	?	00:00:00	[idle_inject/1]
root	17	2	0	Nov15	?	00:00:00	[migration/1]
root	18	2	0	Nov15	?	00:00:00	[ksoftirqd/1]
root	20	2	0	Nov15	?	00:00:00	[kworker/1:0H-kb]
root	21	2	0	Nov15	?	00:00:00	[kdevtmpfs]
root	22	2	0	Nov15	?	00:00:00	[netns]
root	23	2	0	Nov15	?	00:00:00	[rcu_tasks_kthre]

top - 08:12:11 up 2:48, 3 users, load average: 0.74, 0.36, 0.28

Tasks: 295 total, 1 running, 294 sleeping, 0 stopped, 0 zombie

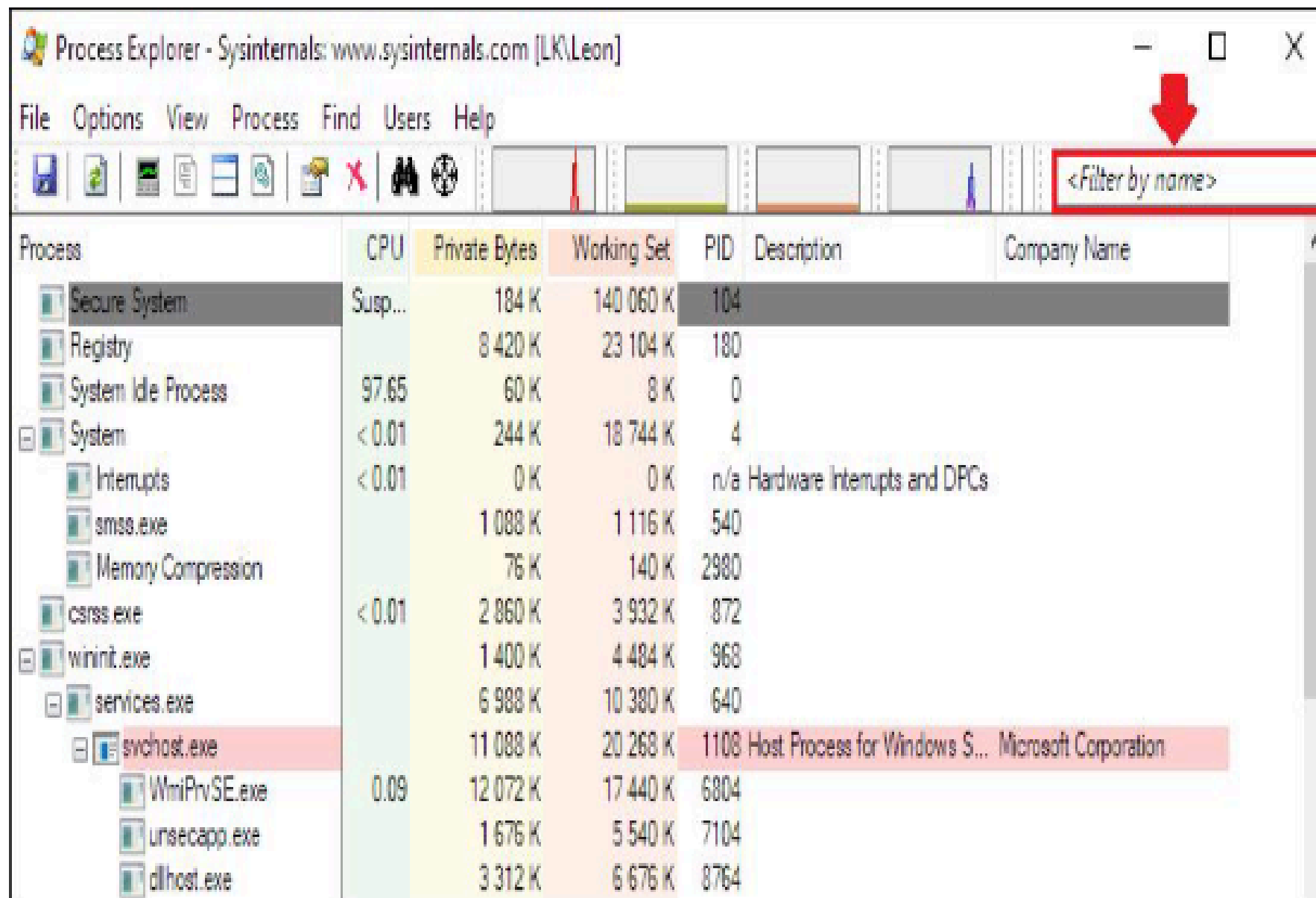
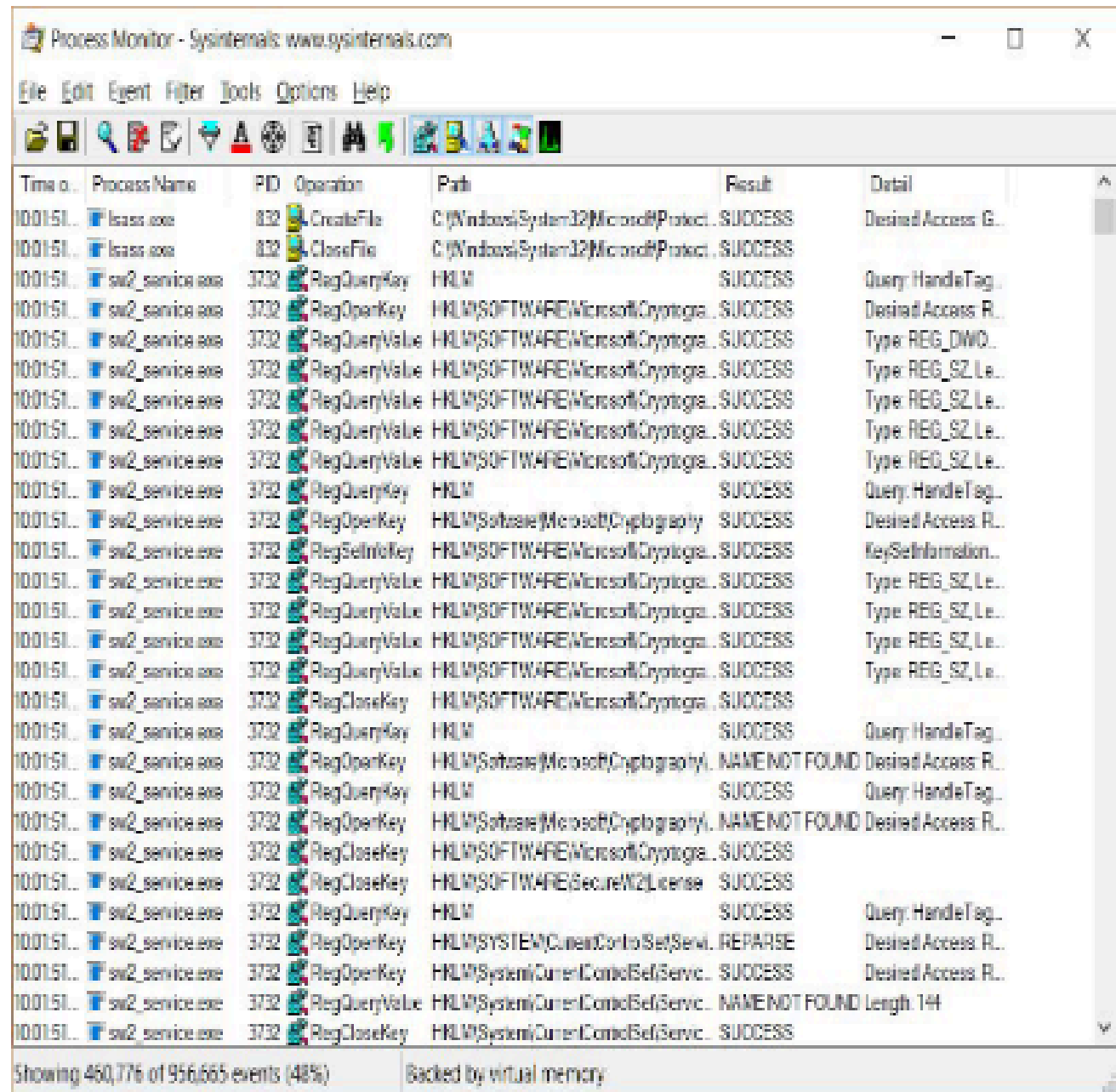
Cpu(s): 5.0 us, 2.3 sy, 0.0 ni, 91.6 id, 0.7 wa, 0.0 hi, 0.3 si, 0.0 st

Mem: 2079.6 total, 138.8 free, 824.1 used, 2006.7 buff/cache

Mem Swap: 947.2 total, 944.2 free, 3.0 used, 1555.3 avail Mem

PID	USER	PR	NI	NT	VSZ	SS	RCPU	MEM	TIME+	COMMAND
652	root	20	0	1158584	36376	15776	S	13.3	1.2	4:12.33 smd
373	root	20	0	0	0	0	S	0.6	0.0	0:15.05 jbd2/ude1-0
574	root	20	0	133700	11572	9020	S	0.6	0.4	0:10.39 netoolad
10	root	20	0	0	0	0	T	0.3	0.0	0:05.63 rcu_sched
3848	ylapirir	20	0	13040	5536	4520	S	0.3	0.2	0:02.61 sshd
33968	ylapirir	20	0	20524	4844	3052	R	0.3	0.1	0:00.28 top
1	root	20	0	166488	18756	7756	S	0.0	0.4	0:07.38 systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.03 kthreadd
3	root	9	-20	0	0	0	T	0.0	0.0	0:00.00 rcu_gp
4	root	9	-20	0	0	0	T	0.0	0.0	0:00.00 rcu_par_gp
6	root	9	-20	0	0	0	T	0.0	0.0	0:00.00 worker/0:0H-kb
8	root	9	-20	0	0	0	T	0.0	0.0	0:00.00 nn_percpu_wq
9	root	20	0	0	0	0	S	0.0	0.0	0:00.63 ksoftirqd/0
11	root	rt	0	0	0	0	S	0.0	0.0	0:00.06 migration/0
12	root	-51	0	0	0	0	S	0.0	0.0	0:00.00 idle_inject/0
14	root	20	0	0	0	0	S	0.0	0.0	0:00.00 cpuhp/0
15	root	20	0	0	0	0	S	0.0	0.0	0:00.00 cpuhp/1

Process Monitoring

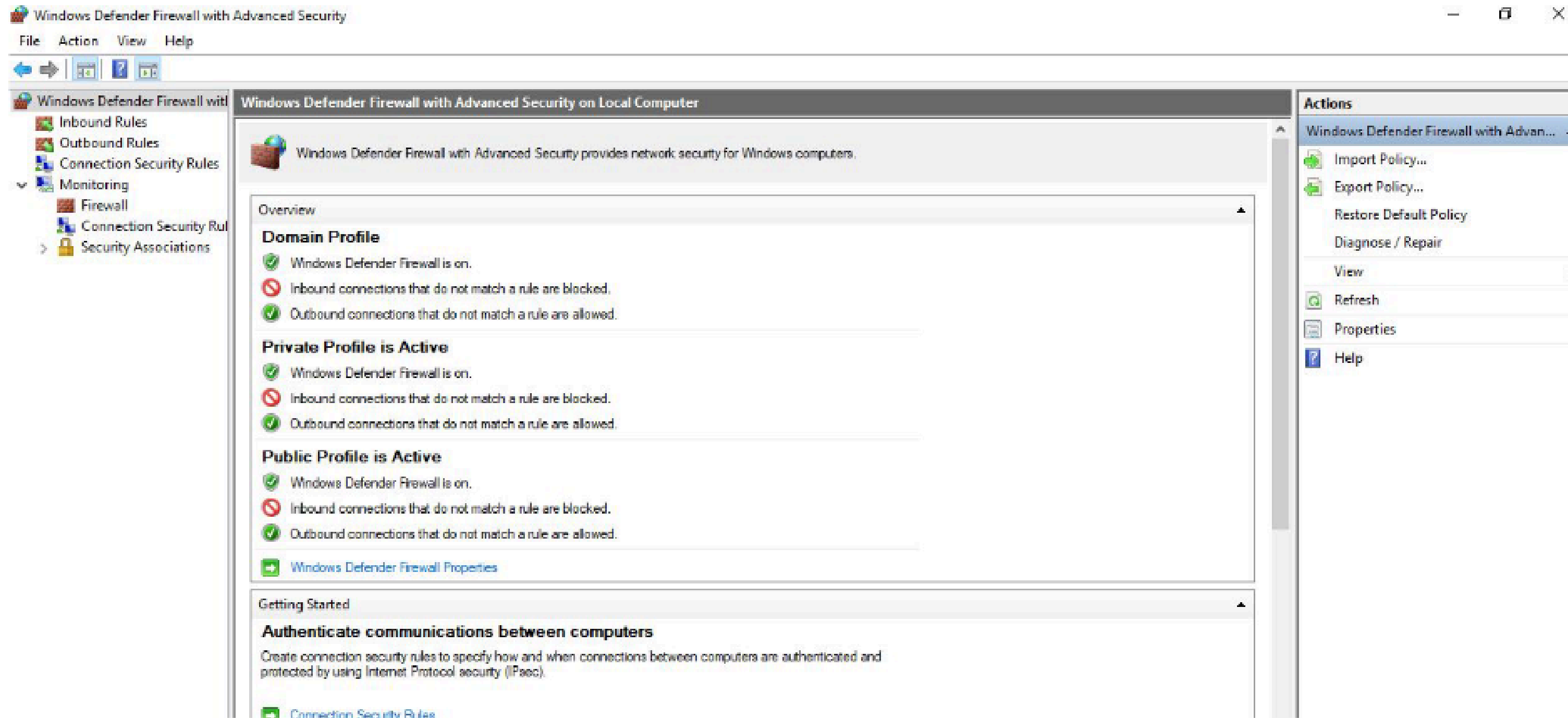


Registry Monitoring

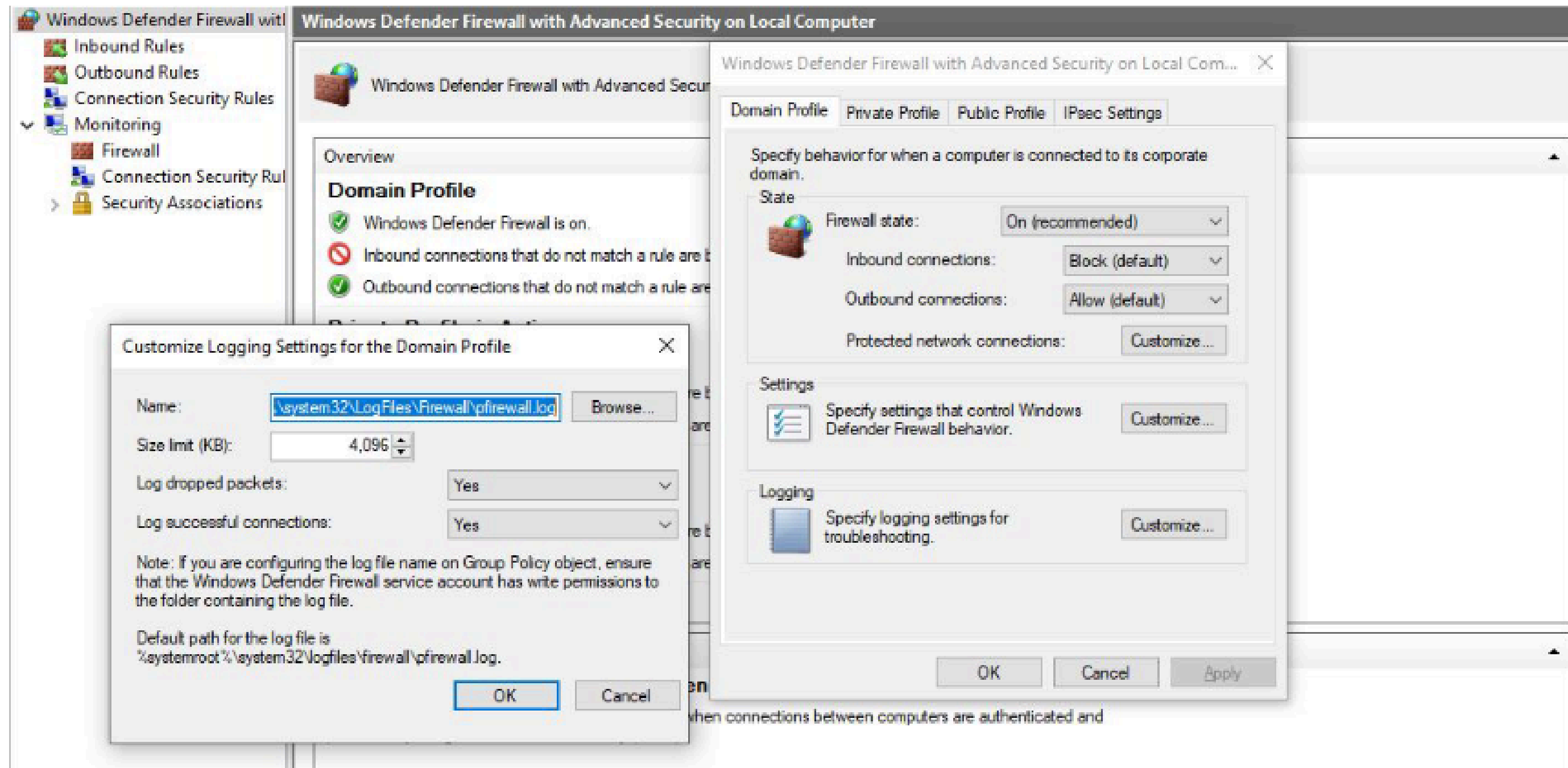
RegShot is a small registry compare utility that allows you to quickly take a snapshot of your registry and then compare it with a second one - done after doing system changes or installing a new software product.



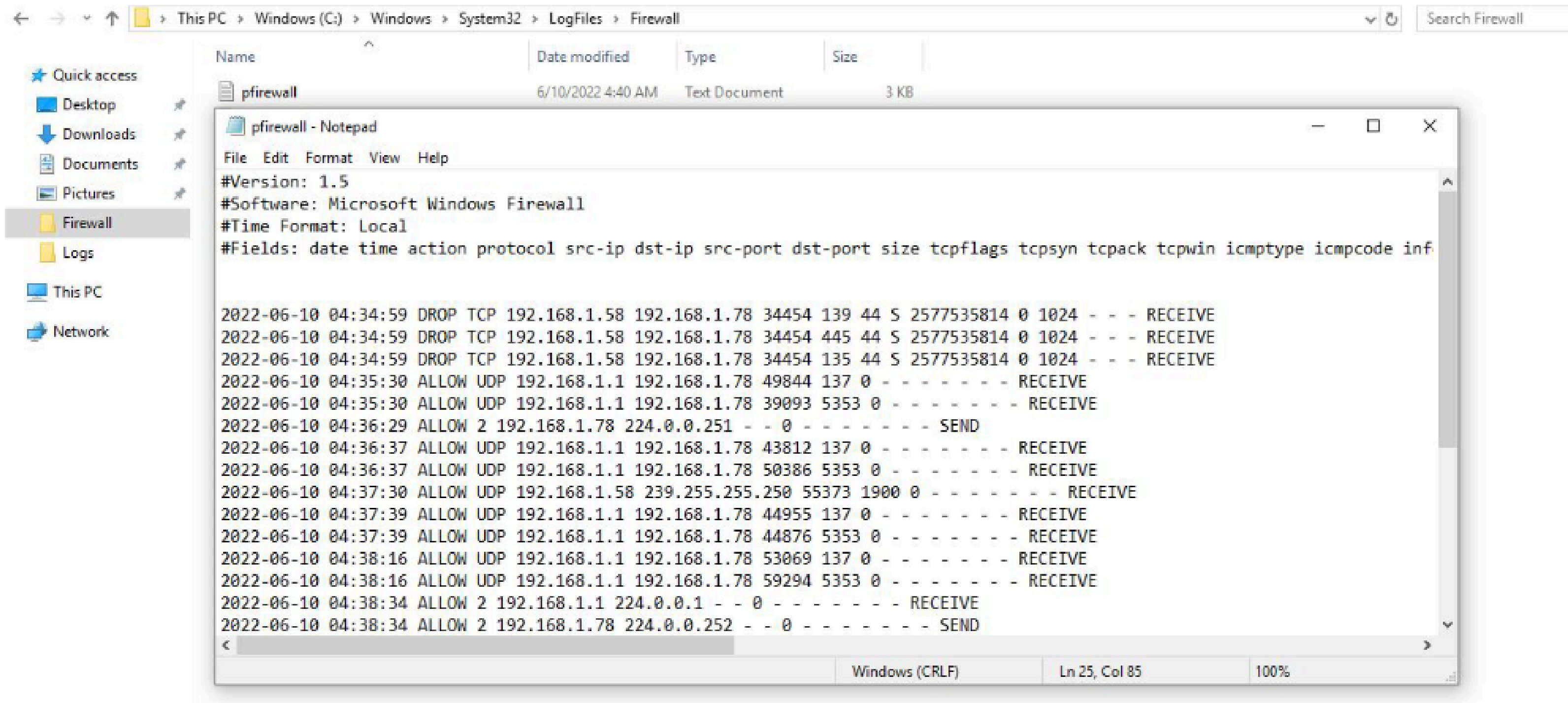
Windows Firewall



Windows Logging



Windows Logging



The screenshot shows a Windows File Explorer window with the address bar set to `This PC > Windows (C:) > Windows > System32 > LogFiles > Firewall`. The file list shows a file named `pfirewall` with a date modified of `6/10/2022 4:40 AM`, type of `Text Document`, and size of `3 KB`. The left sidebar shows the `Firewall` folder selected under `Quick access`. A Notepad window titled `pfirewall - Notepad` is open, displaying the contents of the `pfirewall` file. The text in the Notepad window is as follows:

```
File Edit Format View Help
#Version: 1.5
#Software: Microsoft Windows Firewall
#Time Format: Local
#Fields: date time action protocol src-ip dst-ip src-port dst-port size tcpflags tcpsyn tcpack tcpwin icmptype icmpcode info

2022-06-10 04:34:59 DROP TCP 192.168.1.58 192.168.1.78 34454 139 44 S 2577535814 0 1024 - - - RECEIVE
2022-06-10 04:34:59 DROP TCP 192.168.1.58 192.168.1.78 34454 445 44 S 2577535814 0 1024 - - - RECEIVE
2022-06-10 04:34:59 DROP TCP 192.168.1.58 192.168.1.78 34454 135 44 S 2577535814 0 1024 - - - RECEIVE
2022-06-10 04:35:30 ALLOW UDP 192.168.1.1 192.168.1.78 49844 137 0 - - - - - - - RECEIVE
2022-06-10 04:35:30 ALLOW UDP 192.168.1.1 192.168.1.78 39093 5353 0 - - - - - - - RECEIVE
2022-06-10 04:36:29 ALLOW 2 192.168.1.78 224.0.0.251 - - 0 - - - - - - - SEND
2022-06-10 04:36:37 ALLOW UDP 192.168.1.1 192.168.1.78 43812 137 0 - - - - - - - RECEIVE
2022-06-10 04:36:37 ALLOW UDP 192.168.1.1 192.168.1.78 50386 5353 0 - - - - - - - RECEIVE
2022-06-10 04:37:30 ALLOW UDP 192.168.1.58 239.255.255.250 55373 1900 0 - - - - - - - RECEIVE
2022-06-10 04:37:39 ALLOW UDP 192.168.1.1 192.168.1.78 44955 137 0 - - - - - - - RECEIVE
2022-06-10 04:37:39 ALLOW UDP 192.168.1.1 192.168.1.78 44876 5353 0 - - - - - - - RECEIVE
2022-06-10 04:38:16 ALLOW UDP 192.168.1.1 192.168.1.78 53069 137 0 - - - - - - - RECEIVE
2022-06-10 04:38:16 ALLOW UDP 192.168.1.1 192.168.1.78 59294 5353 0 - - - - - - - RECEIVE
2022-06-10 04:38:34 ALLOW 2 192.168.1.1 224.0.0.1 - - 0 - - - - - - - RECEIVE
2022-06-10 04:38:34 ALLOW 2 192.168.1.78 224.0.0.252 - - 0 - - - - - - - SEND
```

The Notepad window has a status bar at the bottom showing `Windows (CRLF)`, `Ln 25, Col 85`, and `100%`.

EDR Commercial / Open Source



Components of an EDR

- **Data collections agents:**
 - Software components collect data from the endpoint like network activity information, filesystem information, processes running etc.
- **Automated response rules:**
 - Pre-configured rules that identify whether a certain activity is a threat or not and automatically take an action against it.
- **Forensics tools:**
 - Tools that are used by security professionals to investigate incidents or even to perform a threat hunting process.

CB EDR

Carbon Black

C

DASHBOARD

ALERTS

INVESTIGATE

LIVE QUERY

ENFORCE

ENDPOINTS

SETTINGS

ALERT ID: PEWQJW

NON-MALWARE

10:01:20 am Aug 28, 2019

The application powershell.exe attempted to create a memory dump for a system security process (lsass.exe). This may have included user credential or password information. A Terminate Policy Action was applied.

A Administrator

Windows Server 2019 x64

database-layer

Target value

Standard

Quarantine Device

Denied Operation

Terminated

Trusted

Ignored

App Memory

Assessed Target

explorer.exe

powershell.exe

lsass.exe

explorer.exe

Enriched Events

Alert Origin, Behavior, Assets & Tags

ALERT ORIGIN

Primary Process: powershell.exe
Version: LNK632966

The file with hash de79d6e69944c37537dc1ac23833606688fd9f7d7362bf4fe7b1616db32c was detected by Carbon Black.

ALERT BEHAVIORS BASED ON SEVERITY

Data at Risk

Network Threat

Generic Suspect

All Behaviors

run_script, enumerate_processes, break_policy, modify_process, set_system_security, dump_process_memory, read_security_data, policy_terminate

ALERT NOTES & TAGS

Alert PEWQJW was first flagged by CB Defense
10:01:20 am Aug 28, 2019

Dismissed alert PEWQJW on device 11503143
with no comment.
10:01:20 am Aug 28, 2019

Undismissed alert PEWQJW on device
11503143 with no comment.
10:01:20 am Aug 28, 2019

powershell.exe

Policy Action: Terminate

Exception: Trusted White List

Process State: Run

Signature Verification: Not Signed

CMD: "C:\Windows\System32\WindowsPowerShell\Shell.ps1 powershell.exe"

SHA-256: de79d6e69944c37537dc1ac23833606688fd9f7d7362bf4fe7b1616db32c

MD5: 735395b179907ae17c54addcf4239

PID: 3752

Start time: 10:00:36 am Aug 28, 2019

TTPs:

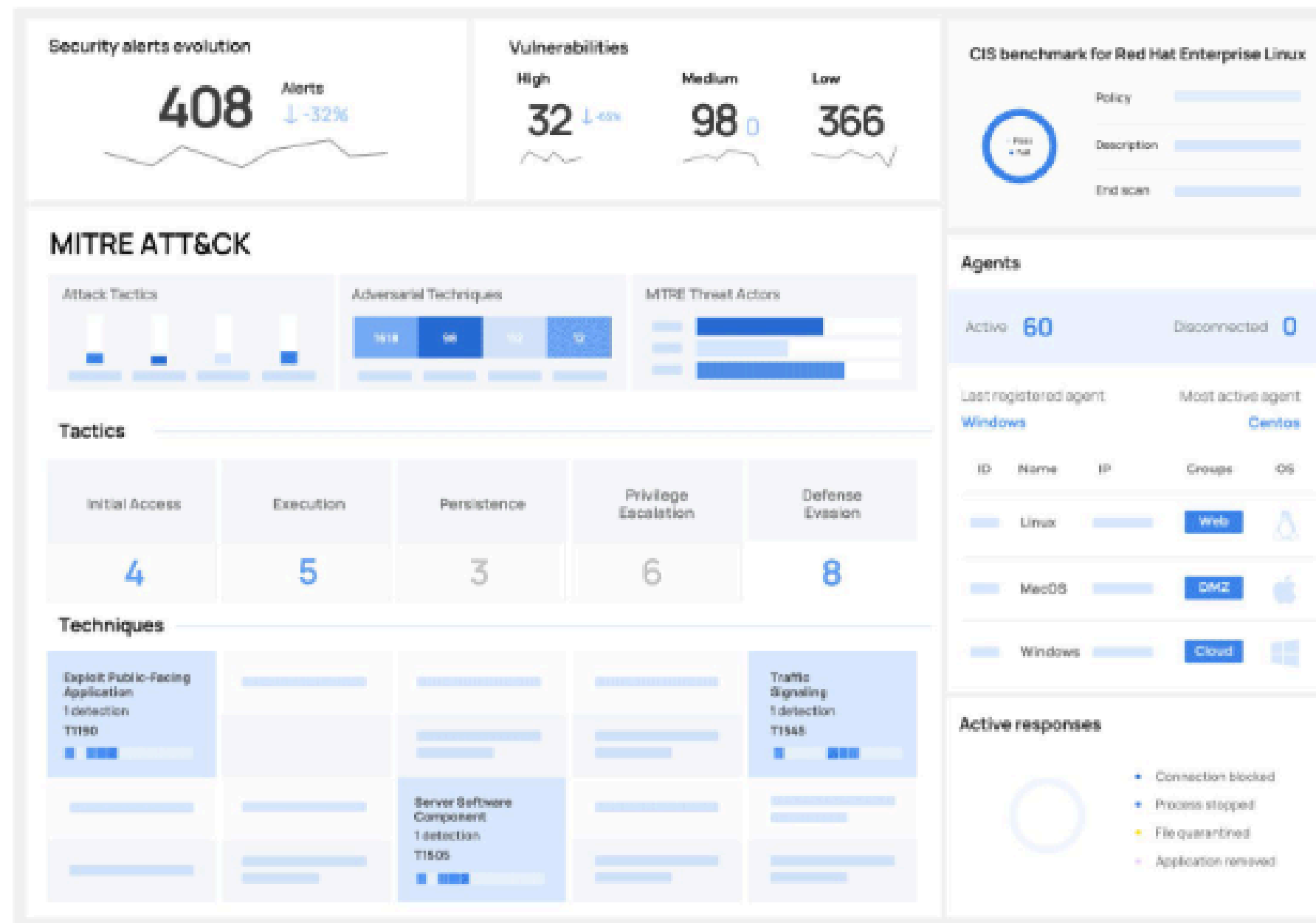
run_script, dump_process_memory, enumerate_processes, set_system_security, read_security_data, policy_terminate, modify_process, bypass_policy

Not Signed

Malware: Not Detected

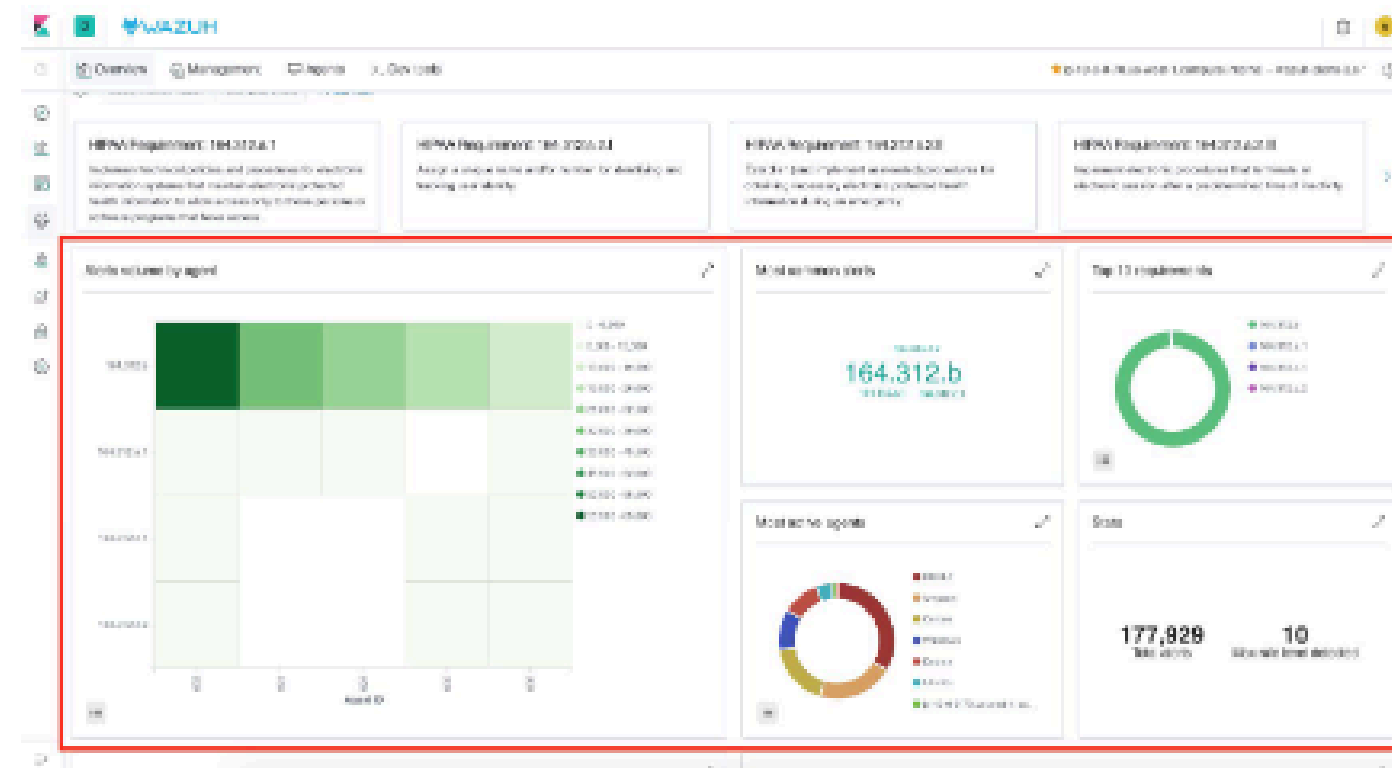
App Origin: The file with hash de79d6e69944c37537dc1ac23833606688fd9f7d7362bf4fe7b1616db32c was detected by Carbon Black.

Wazuh EDR



Wazuh Compliance Scan

- Compliance is defined as being in accordance with certain rules and specifications, and some compliance frameworks state the organisations should meet some security standards.
- For this a compliance scan should be performed also regularly to test the endpoints systems compliance with the security requirements.



Wazuh Integrity Monitoring

